THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 06

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** smallest number expressible as the sum of two cubes in two ways	$1729 = 1^3 + 12^3 = 9^3 + 10^3$
2	**(d)** 625	$25 \times 25 = 625$
3	**(a)** 75000	Move the decimal 4 places right
4	**(a)** Al-Khwarizmi	He wrote on calculation with Hindu numerals
5	**(b)** all sides are equal	Rhombus
6	**(a)** equal and bisect each other	Rectangle property
7	**(d)** Yes	Difference of sums of alternate digits
8	**(c)** 169	$a^2 + 2ab + b^2$
9	**(c)** 9 : 12	Multiply both by 3
10	**(c)** ₹ 300	A gets 5/11 of 660
11	**(d)** 60	12.5/100 × 480
12	**(d)** 1000	800 × 125/100
13	**(b)** 15 cm	$\sqrt{17^2 - 8^2} = 15$
14	**(c)** 6 m	$\sqrt{100 - 64}$
15	**(d)** 6	Workers × days is constant
16	**(a)** cone	Cone
17	**(d)** 5	7 − 2
18	**(c)** 140	$4x + 16$
19	**(c)** 63 cm²	base × height
20	**(d)** 1600 m²	40^2

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	3.5×10^6 (2).	2
22	0 and 1 (1); 101_2 (1).	1+1
23	Arrange digits in descending and ascending order, subtract; repeat to reach 495 (2).	2
24	15 years (2).	2
25	(6, 8, 10), (8, 15, 17); (5, 12, 13) (2).	2
26	$x = 12$, $y = 15$ (2).	2
27	No: mean of 2 and 3 is 2.5 (2).	2
28	Side 13; area 169 cm² (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$1620 = 2^2 \times 3^4 \times 5$ (2); 5 has no pair, so it is not a perfect square (1).	2+1
30	Parallelogram, rectangle, rhombus, square (2); property of parallelograms (1).	2+1
31	Take $a = 5, b = 2$: $9 \neq 21$ (3).	3
32	360 play (1.5); 840 do not (1.5).	1½+1½
33	Packaging boxes, paper models and gift boxes (1.5); designing 3D objects from flat sheets (1.5).	1½+1½
34	Choose equal or closest numbers (1); 11 and 11 (1); product 121 (1).	1+1+1
35	Triplet $(7, 24, 25)$ (1.5); $49 + 576 = 625 = 625$ (1.5).	1½+1½
	OR: By the theorem, height $= \sqrt{41^2 - 9^2}$ (2); $= 40$ m (1).	2+1

36	Area $= \frac{1}{2} \times 30 \times 8 = 120$ m² (2); cost ₹ 1440 (1).	2+1
	OR: Trapezium $= \frac{1}{2}(16 + 24) \times 15 = 300$ cm² (2); rectangle 360 cm² (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $(1 + \frac{1}{4}) \times 4 = 5$ (2). (ii) $x - 4 = 5$, $x = 9$ (3).	2+3
	OR: (i) $2^0 \div 2^3 = 2^{-3} = \frac{1}{8}$ (2). (ii) $\frac{9}{8} \times 6 = \frac{27}{4}$ (3).	2+3
38	(i) CP_1 1000, CP_2 1500; total CP 2500, SP 2400; loss ₹ 100 (3). (ii) Loss percent on a larger base (2).	3+2
	OR: (i) 46000 then 41400 (3). (ii) +3.5% (2).	3+2
39	(i) 400 (1). (ii) 300 (1). (iii) Median (1). (iv) Mean 275 (2).	1+1+1+2
	OR: (i) $(1860 + 1440)/50 = 66$ (3). (ii) Lies between the two means (2).	3+2
40	(i) $\frac{1}{2}ah + \frac{1}{2}bh$ (3). (ii) 300 cm² (2).	3+2
	OR: (i) 3000 m² (2). (ii) ₹ 6000 (1). (iii) 0.3 ha (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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